



ALLAMA IQBAL OPEN UNIVERSITY, ISLAMABAD

Department of Computer Science

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Course Tittle: Database LAB

Course Code: 6907

Semester: 4th

Section: A

Project: Grocery_Store

Instructor Name: Ms. Samreen Akhtar

Project Overview:

- A web-based application for managing grocery store products.
- Built using **Python (Flask)** for backend, **SQL Server** for database, and **HTML/CSS** for frontend.
- Main features:
 - Add new products.
 - Delete existing products.
 - View all products with their categories.

Technology Stack:

Layer	Technology
Backend	Python Flask
Database	SQL Server
Frontend	HTML, CSS, Jinja2

Database Design:

Tables:

1. categories

- category_id → Primary Key, Auto-increment
- category_name → Name of category (e.g., Fruits, Vegetables, Dairy, Snacks)

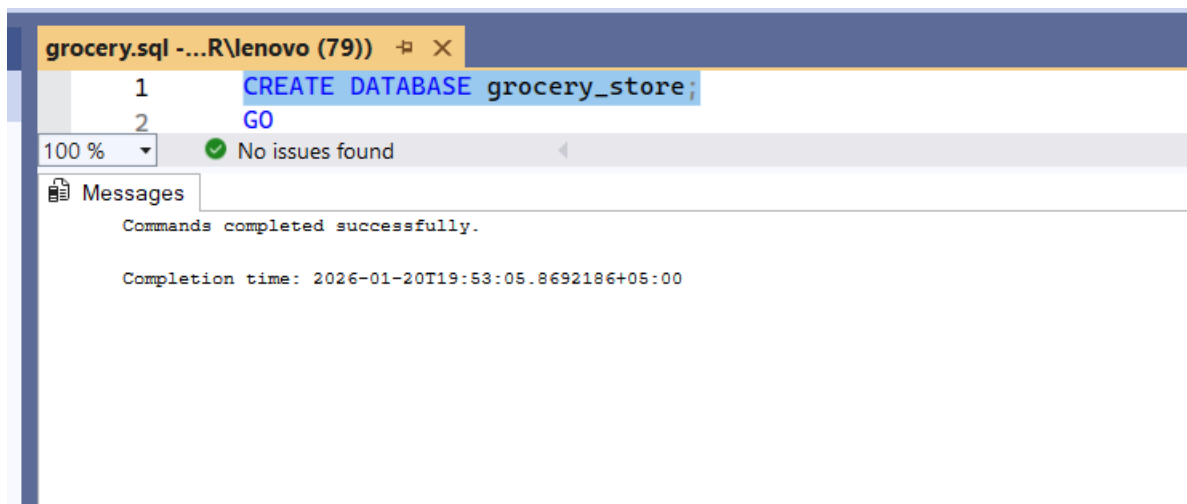
2. products

- product_id → Primary Key, Auto-increment
- product_name → Name of product

- category_id → Foreign Key linking to categories
- price → Product price
- quantity → Available stock
- created_at → Timestamp of when product was added

SQL Code Overview:

Database:



The screenshot shows a SQL script window titled "grocery.sql -...R\lenovo (79)". The script contains two lines: "1 CREATE DATABASE grocery_store;" and "2 GO". The status bar indicates "100 %", "No issues found", and "Commands completed successfully." The completion time is "2026-01-20T19:53:05.8692186+05:00".

```
1 CREATE DATABASE grocery_store;
2 GO
```

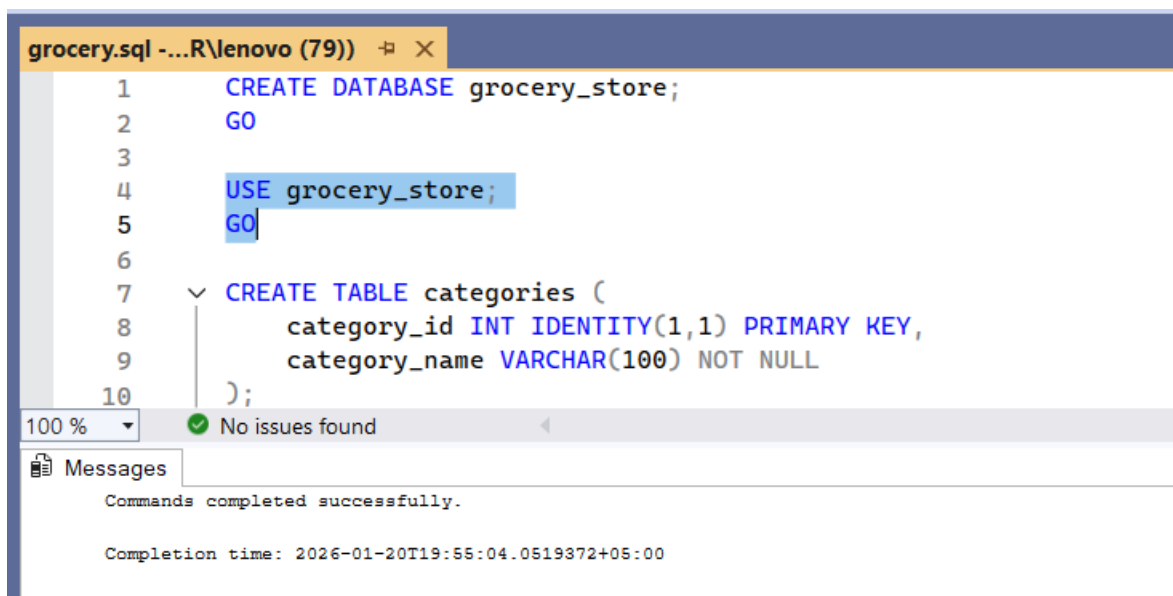
100 % No issues found

Messages

Commands completed successfully.

Completion time: 2026-01-20T19:53:05.8692186+05:00

Use of Database:



The screenshot shows a SQL script window titled "grocery.sql -...R\lenovo (79)". The script contains ten lines: "1 CREATE DATABASE grocery_store;", "2 GO", "3", "4 USE grocery_store;", "5 GO", "6", "7 CREATE TABLE categories (", "8 category_id INT IDENTITY(1,1) PRIMARY KEY,", "9 category_name VARCHAR(100) NOT NULL", "10);". The status bar indicates "100 %", "No issues found", and "Commands completed successfully." The completion time is "2026-01-20T19:55:04.0519372+05:00".

```
1 CREATE DATABASE grocery_store;
2 GO
3
4 USE grocery_store;
5 GO
6
7 CREATE TABLE categories (
8     category_id INT IDENTITY(1,1) PRIMARY KEY,
9     category_name VARCHAR(100) NOT NULL
10 );
```

100 % No issues found

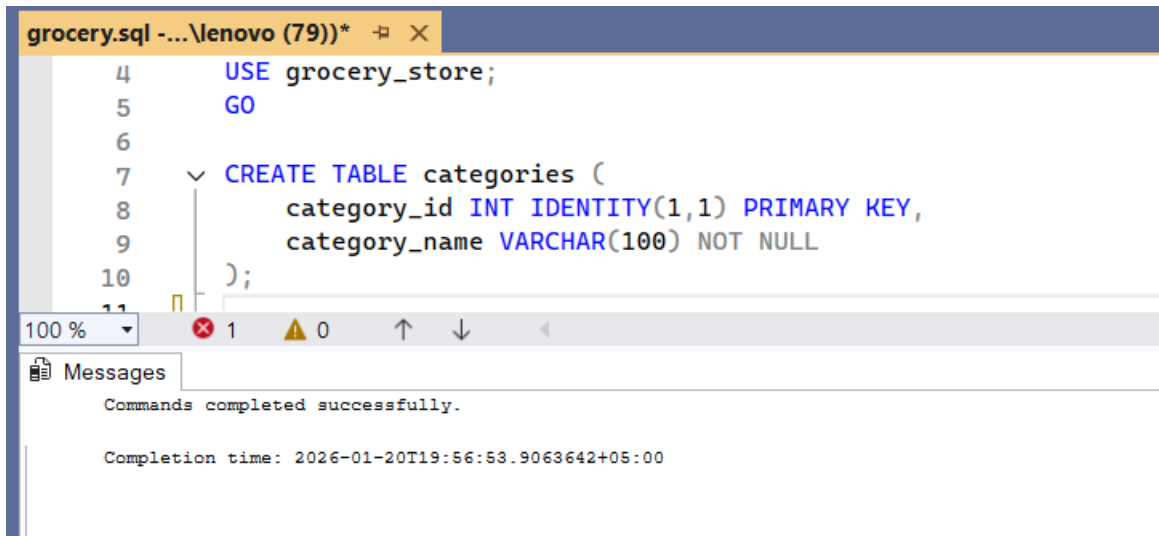
Messages

Commands completed successfully.

Completion time: 2026-01-20T19:55:04.0519372+05:00

Create Table:

1. Categories Table:



The screenshot shows a SQL script window titled "grocery.sql - ...\lenovo (79))*". The script contains the following commands:

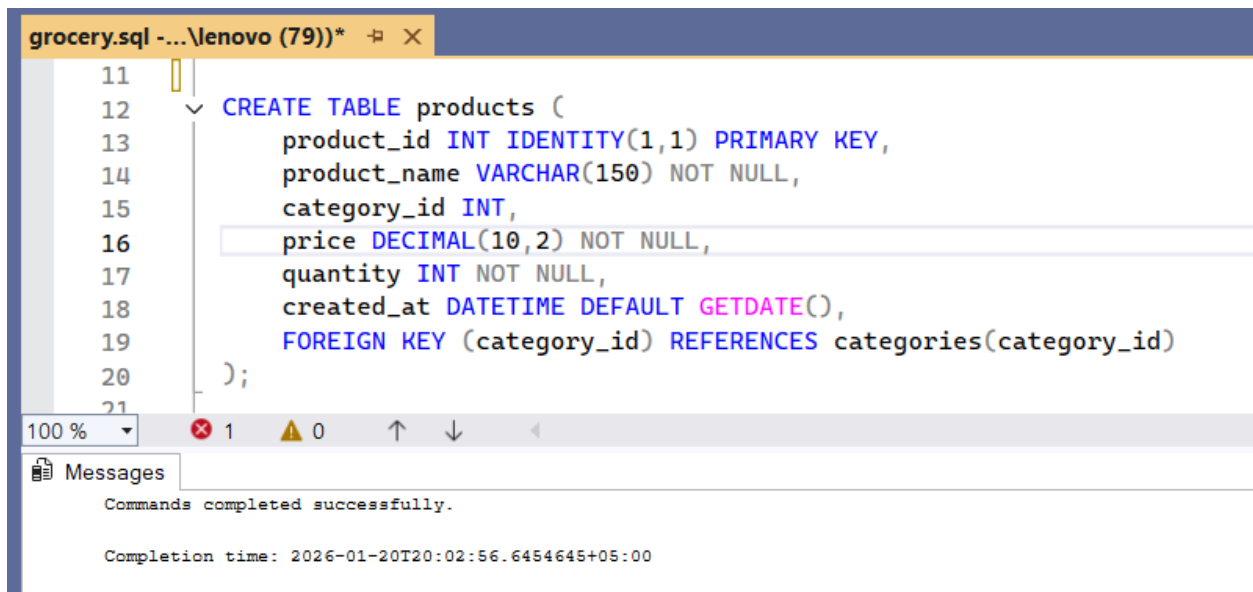
```
4      USE grocery_store;
5      GO
6
7      CREATE TABLE categories (
8          category_id INT IDENTITY(1,1) PRIMARY KEY,
9          category_name VARCHAR(100) NOT NULL
10     );
```

The status bar indicates 1 error and 0 warnings. The Messages pane shows the following output:

```
Commands completed successfully.

Completion time: 2026-01-20T19:56:53.9063642+05:00
```

2. Products Table:



The screenshot shows a SQL script window titled "grocery.sql - ...\lenovo (79))*". The script contains the following commands:

```
11
12     CREATE TABLE products (
13         product_id INT IDENTITY(1,1) PRIMARY KEY,
14         product_name VARCHAR(150) NOT NULL,
15         category_id INT,
16         price DECIMAL(10,2) NOT NULL,
17         quantity INT NOT NULL,
18         created_at DATETIME DEFAULT GETDATE(),
19         FOREIGN KEY (category_id) REFERENCES categories(category_id)
20     );
```

The status bar indicates 1 error and 0 warnings. The Messages pane shows the following output:

```
Commands completed successfully.

Completion time: 2026-01-20T20:02:56.6454645+05:00
```

Insert Values into Tables:

1. Categories:

```
grocery.sql -...\\lenovo (79))*  X
22  INSERT INTO categories (category_name)
23  VALUES ('Fruits'), ('Vegetables'), ('Dairy'), ('Snacks');
24
25  SELECT * FROM categories;
```

100 % 1 0

Results Messages

	category_id	category_name
1	1	Fruits
2	2	Vegetables
3	3	Dairy
4	4	Snacks

2. Products:

```
grocery.sql -...\\lenovo (79))*  X
26
27  INSERT INTO products (product_name, category_id, price, quantity)
28  VALUES
29  ('Apple', 1, 200.00, 50),
30  ('Banana', 1, 150.00, 40),
31  ('Potato', 2, 80.00, 100),
32  ('Milk 1L', 3, 220.00, 30),
33  ('Chips', 4, 100.00, 60);
34  select * from products;
```

100 % 1 0

Results Messages

	product_id	product_name	category_id	price	quantity	created_at
1	1	Apple	1	200.00	50	2026-01-20 20:12:51.190
2	2	Banana	1	150.00	40	2026-01-20 20:12:51.190
3	3	Potato	2	80.00	100	2026-01-20 20:12:51.190
4	4	Milk 1L	3	220.00	30	2026-01-20 20:12:51.190
5	5	Chips	4	100.00	60	2026-01-20 20:12:51.190

Frontend Codes:

1. Html Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Grocery Store</title>
  <style>
    body { font-family: Arial; padding: 20px; }
    table { border-collapse: collapse; width: 100%; }
    th, td { border: 1px solid #ccc; padding: 8px; text-align: center; }
    th { background: #f4f4f4; }
    form { margin-bottom: 20px; }
    input, select, button { padding: 6px; }
  </style>
</head>
<body>

<h2>Add Product</h2>
<form method="POST" action="/add">
  <input type="text" name="name" placeholder="Product Name" required>

  <select name="category" required>
    {% for c in categories %}
      <option value="{{ c.category_id }}">{{ c.category_name }}</option>
    {% endfor %}
  </select>

  <input type="number" name="price" placeholder="Price" required>
  <input type="number" name="quantity" placeholder="Quantity" required>

  <button type="submit">Add Product</button>
</form>

<hr>

<h2>Product List</h2>
<table>
  <tr>
    <th>ID</th>
    <th>Name</th>
    <th>Category</th>
    <th>Price</th>
    <th>Quantity</th>
```

```

        <th>Action</th>
    </tr>

    {% for p in products %}
    <tr>
        <td>{{ p.product_id }}</td>
        <td>{{ p.product_name }}</td>
        <td>{{ p.category_name }}</td>
        <td>{{ p.price }}</td>
        <td>{{ p.quantity }}</td>
        <td>
            <a href="/delete/{{ p.product_id }}">Delete</a>
        </td>
    </tr>
    {% endfor %}
</table>

</body>
</html>

```

2. Style.CSS:

```

* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
    font-family: Arial, Helvetica, sans-serif;
}

body {
    background-color: #f4f6f8;
    padding: 30px;
}

/* Page Title */
h1, h2 {
    text-align: center;
    margin-bottom: 20px;
    color: #333;
}

```

```
/* Form Styling */
form {
    background: #ffffff;
    padding: 20px;
    max-width: 700px;
    margin: 0 auto 30px;
    border-radius: 8px;
    box-shadow: 0 2px 8px rgba(0,0,0,0.1);
}

form input,
form select {
    width: 100%;
    padding: 10px;
    margin-bottom: 12px;
    border: 1px solid #ccc;
    border-radius: 5px;
}

form button {
    background-color: #2ecc71;
    color: #fff;
    border: none;
    padding: 12px;
    width: 100%;
    font-size: 16px;
    border-radius: 5px;
    cursor: pointer;
}

form button:hover {
    background-color: #27ae60;
}

/* Table Styling */
table {
    width: 100%;
    border-collapse: collapse;
    background: #ffffff;
    box-shadow: 0 2px 8px rgba(0,0,0,0.1);
}

table th {
    background-color: #3498db;
```



```
        color: white;
        padding: 12px;
    }

    table td {
        padding: 10px;
        text-align: center;
        border-bottom: 1px solid #ddd;
    }

    table tr:hover {
        background-color: #f1f1f1;
    }

    /* Delete Button */
    a.delete-btn {
        color: white;
        background-color: #e74c3c;
        padding: 6px 10px;
        text-decoration: none;
        border-radius: 4px;
    }

    a.delete-btn:hover {
        background-color: #c0392b;
    }

    /* Responsive */
    @media (max-width: 768px) {
        table, thead, tbody, th, td, tr {
            display: block;
        }

        table tr {
            margin-bottom: 15px;
        }

        table td {
            text-align: right;
            padding-left: 50%;
            position: relative;
        }

        table td::before {
            content: attr(data-label);
        }
    }
}
```

```
    position: absolute;
    left: 10px;
    width: 45%;
    padding-left: 10px;
    font-weight: bold;
    text-align: left;
  }
}
```

3. Script.js:

```
document.addEventListener("DOMContentLoaded", function () {

    const deleteButtons = document.querySelectorAll(".delete-btn");

    deleteButtons.forEach(function (btn) {
        btn.addEventListener("click", function (event) {
            if (!confirm("Are you sure you want to delete this product?")) {
                event.preventDefault();
            }
        });
    });

    const form = document.querySelector("form");

    form.addEventListener("submit", function (event) {
        const price = document.querySelector("input[name='price']").value;
        const quantity = document.querySelector("input[name='quantity']").value;

        if (price <= 0 || quantity <= 0) {
            alert("Price & Quantity must be greater than zero");
            event.preventDefault();
        }
    });
});
```

Output:

Grocery Store

web.whatsapp.com

C:/Users/lenovo/OneDrive/Desktop/grocery_store/frontend/dynamic/index.html

Add Product

Product Name

{{ c.category_name }}

Price

Quantity

Add Product

Product List

{% for p in products %}{% endfor %}

ID	Name	Category	Price	Quantity	Action
{{ p.product_id }}	{{ p.product_name }}	{{ p.category_name }}	{{ p.price }}	{{ p.quantity }}	Delete

53°F
Partly cloudy

Search

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UK

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